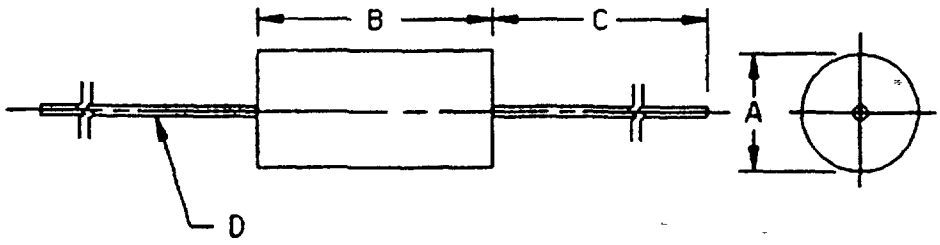
INCH-POUND
MIL-C-39010/30
4 January 1994
SUPERSEDING
MIL-C-39010/30
1 March 1985

MILITARY SPECIFICATION SHEET

COILS, FIXED, RADIOFREQUENCY, MOLDED, MICROMINIATURE, MAGNETICALLY SHIELDED (FERRITE CORE-FERRITE SLEEVE), ESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39010.



Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
A	.152 (3.86)	172 (4 37)
B	.390 (9.91)	430 (10 92)
C	1.250 (31.75)	1 626 (41 30)
D	023 (0 58)	027 (0 69)

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for information only.

FIGURE 1 Dimensions and configuration

(D) denotes changes

AMSC N/A
FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1

Material: Ferrite core with ferrite sleeve

(D) Weight. 0.04 ounce maximum

Operating temperature range -55°C to +105°C.

Dielectric withstanding voltage. Method 301 of MIL-STD-202; test voltage of 1,000 V rms

Barometric pressure Method 105, test condition C, of MIL-STD-202, 70,000 feet with a test voltage of 200 V rms

Electrical characteristics See tables I and II.

Inductance: See table I.

Inductance tolerance See table I.

Q values See table I

Self-resonant frequency See table I

Percent coupling 3 percent maximum

DC resistance: See table I

Temperature rise 15°C.

Terminal pull: 5 pounds

(D) Part or Identifying Number (PIN) M39010/03 (dash number from table I).

(D) Supersession data This specification supersedes MS75089 dated 4 September 1985 and MS21363 dated 2 July 1985

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TABLE I. Electrical characteristics (initial) and dash numbers

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/	Incremental current
A150**	15	5, 10	45	2.5	49	.80	315.0	250
A160**	16	5	45	2.5	45	.89	300 0	235
A180**	18	5, 10	45	2.5	45	.89	300 0	235
A200**	20	5	45	2.5	41	96	290.0	220
A220**	22	5, 10	45	2.5	41	96	290 0	220
A240**	24	5	45	2.5	38	1.19	260.0	200
A270**	27	5, 10	45	2.5	38	1.19	260.0	200
A300**	30	5	45	2.5	34	1.37	240.0	190
A330**	33	5, 10	45	2.5	34	1.37	240.0	190
A360**	36	5	45	2.5	29	1.93	205.0	180
A390**	39	5, 10	50	2.5	29	1 93	205.0	180
A430**	43	5	50	2.5	27	2 11	195 0	175
A470**	47	5, 10	50	2 5	27	2 11	195.0	175
A510**	51	5	50	2.5	25	2 23	190 0	160
A560**	56	5, 10	50	2.5	25	2.23	190.0	160
A620**	62	5	50	2.5	21	2.44	180.0	150
A680**	68	5, 10	50	2.5	21	2.44	180.0	150
A750**	75	5	50	2 5	10 5	2.70	170.0	140
A820**	82	5, 10	50	2 5	10 5	2 70	170 0	140
A910**	91	5	50	2.5	10.0	3 12	160 0	120
A101**	100	5, 10	50	2.5	10 0	3.12	160.0	120
A111**	110	5	50	.79	9.7	3.60	150 0	95
A121**	120	5, 10	55	.79	9 7	3.60	150 0	95
A131**	130	5	55	79	8.5	4.10	140 0	90
A151**	150	5, 10	55	79	8.5	4.10	140.0	90
A161**	160	5	55	79	8.0	4.40	135.0	85
A181**	180	5, 10	55	79	8.0	4 40	135.0	85
A201**	200	5	55	79	7.5	5 00	125.0	80
A221**	220	5, 10	55	79	7 5	5.00	125.0	80
A241**	240	5	55	.79	7 0	5.80	115.0	70
A271**	270	5, 10	55	79	7 0	5.80	115.0	70
A301**	300	5	55	79	6 5	6.40	110.0	65

See footnotes at end of table

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TABLE I Electrical characteristics (initial) end-dash-numbers - Continued

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/	Incremental current
A331**	330	5, 10	55	.79	6.5	6.40	110.0	65
A361**	360	5	55	.79	6 2	7.40	105.0	60
A391**	390	5, 10	60	.79	6.2	7.40	105.0	60
A431**	430	5	60	.79	5.7	9.50	92 0	58
A471**	470	5, 10	60	79	5.7	9.50	92.0	58
A511**	510	5	60	.79	4 7	10.5	90.0	55
A561**	560	5, 10	60	.79	4.7	10.5	90.0	55
A621**	620	5	60	.79	4.5	11.8	80.0	50
A681**	680	5, 10	60	.79	4.5	11.8	80.0	50
A751**	750	5	60	79	4.2	13.0	80.0	45
A821**	820	5, 10	60	79	4.2	13.0	80.0	45
A911**	910	5	60	.79	3.8	17.5	70.0	40
A102**	1,000	5, 10	60	79	3.8	17 5	70 0	40
A112**	1,100	5	45	.25	1.5	22 1	60 0	35
A122**	1,200	5, 10	45	.25	1.5	22.1	60.0	35
A132**	1,300	5	45	.25	1.2	26.5	55.0	33
A152**	1,500	5, 10	45	25	1.2	26.5	55.0	33
A162**	1,600	5	45	.25	1.0	29 9	50.0	30
A182**	1,800	5, 10	45	.25	1.0	29.9	50.0	30
A202**	2,000	5	45	25	.97	33 8	50 0	27
A222**	2,200	5, 10	45	.25	97	33.8	50 0	27
A242**	2,400	5	45	.25	92	47.3	40 0	25
A272**	2,700	5, 10	45	25	92	47.3	40 0	25
A302**	3,000	5	45	.25	84	53 0	40 0	22
A332**	3,300	5, 10	45	.25	.84	53.0	40.0	22
A362**	3,600	5	45	.25	.80	73.8	35 0	20
A392**	3,900	5, 10	45	.25	80	73 8	35 0	20
A432**	4,300	5	45	.25	.74	81 6	31 0	19
A472**	4,700	5, 10	45	.25	74	81 6	31 0	19
A512**	5,100	5	44	25	.73	98 9	28 0	17
A562**	5,600	5, 10	44	25	73	98 9	28 0	17
A622**	6,200	5	40	.25	66	111 0	27 0	16

See footnotes at end of table

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TABLE I. Electrical characteristics (initial) and dash numbers - Continued

Dash number 1/	Inductance µH	Inductance tolerance ± percent	Q min	Test frequency (MHz)	Self-resonant frequency minimum (MHz)	DC resistance (25°C) maximum (ohms)	Rated dc current (mA) 2/	Incremental current
A682**	6,800	5, 10	40	.25	.66	111.0	27.0	16
A752**	7,500	5	40	.25	.54	119.0	26.0	15
A822**	8,200	5, 10	40	.25	.54	119 0	26.0	15
A912**	9,100	5	40	.25	.47	137.0	24.0	14
A103**	10,000	5, 10	40	.25	.47	137.0	24.0	14
A113**	11,000	5	30	.079	.33	143.0	23 0	13
A123**	12,000	5, 10	30	.079	.33	143.0	23.0	13
A133**	13,000	5	30	.079	.29	157 0	22.0	12
A153**	15,000	5, 10	30	.079	.29	157 0	22 0	12
A163**	16,000	5	30	079	.28	175.0	21.0	10
A183**	18,000	5, 10	30	.079	.28	175 0	21.0	10
A203**	20,000	5	27	.079	.25	274.0	17 0	9
A223**	22,000	5, 10	27	.079	.25	274.0	17.0	9
A243**	24,000	5	27	079	.21	308 0	16 0	8
A273**	27,000	5, 10	27	.079	.21	308 0	16.0	8
A303**	30,000	5	27	.079	.19	343.0	15.0	7 5
A333**	33,000	5, 10	27	.079	.19	343.0	15 0	7.5
A363**	36,000	5	27	.079	.17	376.0	15.0	6.0
A393**	39,000	5, 10	27	.079	.17	376 0	15.0	6.0
A433**	43,000	5	23	.079	.16	473.0	13.0	5.5
A473**	47,000	5, 10	23	079	16	473.0	13 0	5.5
A513**	51,000	5	23	.079	.14	512.0	13 0	5.0
A563**	56,000	5, 10	23	.079	14	512.0	13.0	5.0
A623**	62,000	5	23	079	.13	580 0	12.0	4.0
A683**	68,000	5, 10	23	.079	.13	580 0	12.0	4.0
A753**	75,000	5	21	.079	12	618 0	11 0	3.5
A823**	82,000	5, 10	21	.079	.12	618 0	11.0	3 5
A913**	91,000	5	18	.079	11	678 0	11 0	3 0
A104**	100,000	5, 10	18	.079	11	678.0	11.0	3.0

1/ The complete dash number will include two additional letters (indicated by **) The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the failure rate symbol (e g , M, P, R, S) and will be added to the end of the dash number

2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.

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(D) TABLE II. Electrical characteristics (final) 1/

Inspection group	Allowable variation from initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5	---	---	-10
Group IV	±10	±(5% + 001 ohm)	2/	-20
Group V	±5	±(2% + 001 ohm)	2/	-20
Quality conformance inspection Group B				
Subgroup 1	±5	±(2% + 001 ohm)	2/	-20
Subgroup 3	±5	---	---	-10
Subgroup 4	±10	±(5% + 001 ohm)	2/	-20

1/ Test fixture allowance of + 01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).

2/ The self-resonant frequency shall be not less than the value specified in table I

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR, ME, MI
Navy - AS, CG, MC, OS, SH
Air Force - 17, 19, 99
DLA - ES

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5950-0831-03)

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